

3-D Point Cloud Classification using Traditional vs. AIGC-based Augmentation

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Abstract—In recent years, 3-D point cloud classification has emerged as a critical task in computer vision, robotics, and autonomous systems. While deep learning architectures like PointNet have shown promising results in modeling unordered point sets, their performance heavily relies on data diversity and robustness. To address this, data augmentation plays a pivotal role in enhancing model generalization. In this study, we systematically evaluate and compare two augmentation paradigms for 3-D point clouds: traditional geometric transformations (e.g., rotation, scaling, jittering) and generative data augmentation based on AIGC (AI-Generated Content), specifically using Point-E, a diffusion-model-based point cloud synthesizer.

We conduct six experiments on the ModelNet40 dataset using the PointNet architecture. These experiments include no augmentation (baseline), traditional augmentation, AIGC-based augmentation, and various hybrid configurations. Each trained model is evaluated on a standardized test set containing 1000 samples across 40 object classes, and the evaluation metrics include test accuracy and F1-score. Our findings reveal that traditional augmentation significantly improves classification performance over the baseline, whereas the AIGC-based augmentation, when used alone, leads to substantial performance degradation. However, hybrid strategies that combine baseline data with traditional augmentations yield the highest accuracy, outperforming other configurations.

These results highlight the reliability and effectiveness of conventional augmentations in 3-D deep learning tasks and suggest that current generative augmentation methods require further refinement to match or exceed traditional strategies. This work provides insights into the trade-offs and potential of different augmentation approaches for enhancing 3-D point cloud classifiers.

Keywords—3-D point cloud classification, AIGC, generative models, Point-E, deep learning, model generalization.

I. INTRODUCTION

THREE-dimensional (3-D) point cloud data has become increasingly central to a wide range of applications, including autonomous driving, robotic navigation, augmented reality, and 3-D scene understanding. Unlike images, point clouds are unordered sets of spatial points in Euclidean space, typically denoted as $P = \{p_i = (x_i, y_i, z_i)\}_{i=1}^N$ where N is the number of points. Due to the absence of grid structure and inherent permutation invariance, traditional convolutional neural networks (CNNs) are not directly applicable to such data. PointNet, a pioneering architecture proposed by Qi *et al.* [1], addressed these challenges by introducing a symmetric function-based learning strategy, enabling effective deep learning on raw point clouds.

Despite its innovation, PointNet and its successors suffer from limited generalization when exposed to data distributions that differ from the training set. This is particularly problematic in real-world scenarios where 3-D data is noisy, occluded, or observed from novel perspectives. To alleviate these challenges, data augmentation has proven to be an essential technique. Traditional augmentation strategies, such as random rotation, scaling, and jittering, introduce geometric variability and have been widely adopted to improve robustness in point cloud classification tasks [2].

In parallel, the emergence of generative models has opened new avenues for data augmentation. Diffusion-based methods such as Point-E [3] enable the synthesis of realistic point clouds by learning data distributions through iterative denoising processes. These generative approaches fall under the broader category of AI-generated content (AIGC), offering potential for enriching training datasets with diverse and unseen shapes. However, their actual impact on classification performance remains insufficiently studied.

This work aims to conduct a systematic comparison between traditional and AIGC-based data augmentation methods using the PointNet architecture on the ModelNet40 benchmark. We implement six experimental configurations—ranging from raw data only to hybrid augmentation strategies—and evaluate their effect on model generalization. Through comprehensive analysis, we highlight the strengths and limitations of each approach and provide insight into the current viability of generative augmentation in 3-D point cloud learning.

II. RELATED WORK

The problem of 3-D point cloud classification has attracted extensive attention in recent years, driven by advances in deep learning and increasing demand for 3-D perception in real-world systems. Early methods attempted to project 3-D point clouds into 2-D views [4] or voxelize them into volumetric grids [5]. While these techniques enable the use of traditional convolutional networks, they suffer from significant limitations, such as information loss due to quantization and high computational cost.

To address these challenges, Qi *et al.* introduced PointNet [1], which directly processes raw point sets using shared multilayer perceptrons (MLPs) and a symmetric max-pooling function to achieve permutation invariance. This architecture preserves the full geometric information and serves as the backbone for many subsequent works. Its improved version, PointNet++ [6], incorporates local region hierarchies using neighborhood sampling and grouping, further enhancing the

capacity to capture fine-grained features.

Data augmentation has long been a staple for improving generalization in point cloud learning. Traditional methods include geometric transformations such as random rotation, translation, scaling, jittering, and point dropout [7]. These augmentations inject structural variability and prevent overfitting, especially in small or imbalanced datasets.

More recently, generative models have been explored for data augmentation. Generative Adversarial Networks (GANs) [8], Variational Autoencoders (VAEs) [9], and diffusion models have demonstrated potential in synthesizing plausible 3-D structures. PointFlow [10] and ShapeGF [11] represent early attempts to generate point clouds via continuous distribution modeling. Point-E [3], a diffusion-based model, represents a significant step forward by enabling 3-D point cloud generation directly from text or image prompts. However, the application of these AIGC-based models for augmenting training datasets in supervised classification tasks is still in its infancy, and their comparative effectiveness against traditional methods remains underexplored.

This study aims to bridge that gap by empirically comparing traditional augmentation with AIGC-generated augmentation using Point-E and evaluating their impact on PointNet-based classification performance.

III. THEORETICAL FOUNDATIONS

Point cloud classification presents unique challenges due to the unstructured, unordered, and irregular nature of 3-D data. Each point cloud is typically represented as a set of points $\{p_1, p_2, \dots, p_N\}$, where each point $p_i \in \mathbb{R}^3$ describes its spatial coordinates. Unlike images or voxel grids, point clouds do not possess a regular topology, making traditional convolutional neural networks (CNNs) inapplicable. To address this, PointNet [1] was proposed as a pioneering architecture that directly operates on raw point sets while preserving permutation invariance.

A. PointNet Architecture

The PointNet model is designed around three core principles: transformation invariance, point-wise feature extraction, and symmetric aggregation. The input point cloud is first processed through a transformation network, referred to as T-Net, which learns an affine transformation matrix. This transformation maps each input point p_i to a canonical space by computing $\tilde{p}_i = T_1 \cdot p_i$, where $T_1 \in \mathbb{R}^{3 \times 3}$ is the predicted matrix and $\tilde{p}_i$ is the transformed point.

Following this, a set of shared multilayer perceptrons (MLPs) is applied to each point independently, producing point-wise features. The network learns a mapping $h_i = \phi(\tilde{p}_i)$, where $h_i \in \mathbb{R}^d$ is a high-dimensional feature vector and $\phi(\cdot)$ denotes the nonlinear transformation function consisting of MLP layers. These operations are shared across all NNN points, which preserves the model's invariance to the ordering of input data.

To further enhance the robustness of feature representations, a second transformation network is used to apply a learned transformation $T_2 \in \mathbb{R}^{d \times d}$ to the features h_i , resulting in aligned features $h'_i = T_2 \cdot h_i$. To ensure this matrix remains

close to orthogonal and does not introduce distortions, a regularization term is added to the loss function. This term is defined as

$$\mathcal{L}_{reg} = \|I - T_2 \cdot T_2^T\|_F^2 \quad (1)$$

where I is the identity matrix and $\|\cdot\|$ denotes the Frobenius norm.

After transformation, the point-wise features are aggregated using a symmetric function, specifically max pooling, to produce a global feature vector that summarizes the entire shape. This operation is expressed as

$$g = \max(h'_1, h'_2, \dots, h'_N) \quad (2)$$

where $g \in \mathbb{R}^d$ is the global descriptor of the object.

The global feature vector is then passed through a series of fully connected layers and a final softmax classifier to generate the class prediction vector $\hat{y}$. The primary learning objective is to minimize the cross-entropy loss between the predicted output and the ground truth label y , which is defined as

$$\mathcal{L}_{CE} = - \sum_{c=1}^C y_c \log(\hat{y}_c) \quad (3)$$

where C is the number of classes. The total loss function used for training the model is the sum of the classification and regularization terms:

$$\mathcal{L}_{total} = \mathcal{L}_{CE} + \lambda \cdot \mathcal{L}_{reg} \quad (4)$$

where λ is a hyperparameter controlling the strength of the regularization.

B. Traditional Data Augmentation

To improve generalization, traditional data augmentation techniques are commonly used during training. These techniques include spatial transformations such as rotation, scaling, jittering, and point dropout, which simulate geometric variations observed in real-world data. A commonly used rotation transformation is applied around the vertical (Y) axis. The rotation of a point $p_i = (x_i, y_i, z_i)$ by an angle θ is defined by

$$R(\theta) = \begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix}, p'_i = R(\theta) \cdot p_i \quad (5)$$

Scaling is applied by multiplying all coordinates of a point by a random scalar s , sampled from a predefined interval, typically $s \in [0.8, 1.2]$, resulting in

$$p'_i = s \cdot p_i \quad (6)$$

Jittering simulates sensor noise by adding Gaussian noise to each point:

$$p'_i = p_i + \epsilon, \epsilon \sim \mathcal{N}(0, \sigma^2) \quad (7)$$

Point dropout randomly removes a subset of points from the input, helping the network generalize to incomplete or occluded inputs.

These augmentations are label-preserving and can be applied independently or in combination. Their computational simplicity and efficacy have made them standard practices in 3-D learning pipelines.

C. Generative Augmentation via Diffusion Models

While traditional methods rely on manual transformations, recent work in AI-generated content (AIGC) proposes the use of generative models to synthesize entirely new point clouds.

Point-E [3] is a diffusion-based model capable of generating 3-D shapes from textual or visual prompts. It operates in two phases: a forward noising process and a learned reverse denoising process.

In the forward process, a clean point cloud x_0 is progressively corrupted by Gaussian noise according to the rule

$$x_t = \sqrt{\alpha_t} \cdot x_0 + \sqrt{1 - \alpha_t} \cdot \epsilon, \epsilon \sim \mathcal{N}(0, I) \quad (8)$$

where α_t is a predefined noise schedule at step t . The reverse process attempts to reconstruct x_0 by estimating the added noise ϵ using a neural network $\epsilon_\theta(x_t, t)$. The denoising step is computed as

$$x_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{1 - \alpha_t}{\sqrt{1 - \alpha_t}} \cdot \epsilon_\theta(x_t, t) \right) + \sigma_t \cdot z \quad (9)$$

where $z \sim \mathcal{N}(0, I)$, $\bar{\alpha}_t$ is the cumulative noise product, and σ_t controls the stochasticity of the denoising process.

Point clouds generated through this process can be conditionally sampled for specific classes and used as additional training data. However, due to variability in quality, structure, and semantic consistency, their effectiveness as augmentations remains an open empirical question.

This work provides a direct comparison between traditional augmentation and Point-E-based generative augmentation, assessing their impact on 3-D classification performance in a unified experimental framework.

IV. METHODS

This section describes the complete experimental pipeline designed to evaluate the impact of traditional and AIGC-based augmentation on 3-D point cloud classification performance. The workflow comprises six components: (i) construction of a standardized test dataset, (ii) training set generation under different augmentation protocols, (iii) implementation of the PointNet classification model, (iv) supervised training across augmentation conditions, (v) test-time evaluation, and (vi) statistical and visual analysis.

A. Test Dataset Construction

To provide a consistent benchmark for evaluating all experimental configurations, a fixed test dataset was constructed by uniformly sampling 25 objects from each of the 40 categories in the ModelNet40 benchmark. This process, implemented in a dedicated script, involved parsing and extracting samples from the original HDF5-formatted test files provided with the dataset. Each selected point cloud, originally containing 2048 points, was first downsampled to 1024 points using uniform sampling without replacement.

Subsequently, each sample was subjected to spatial normalization. The centroid of each point cloud was aligned to the origin by subtracting the mean coordinate. Then, the set of points was scaled such that the maximum Euclidean distance from the origin did not exceed one. Formally, for a point cloud $P = \{p_i\}_{i=1}^N$, the transformation is given by:

$$\bar{p} = \frac{1}{N} \sum_{i=1}^N p_i, \hat{p}_i = \frac{p_i - \bar{p}}{\max_j \|p_j - \bar{p}\|^2} \quad (10)$$

This guarantees that all test data reside within a unit sphere centered at the origin, with uniform point count and zero mean.

B. Training Dataset and Augmentation Strategies

Six experimental conditions were implemented to investigate the effects of data augmentation strategies. In each case, the training dataset consisted of 40 class folders, each containing 25 point clouds. Three augmentation conditions were considered: no augmentation (Exp1), traditional geometric augmentation (Exp2), and generative augmentation via a diffusion-based model (Exp3). Additionally, combinations of these conditions were examined (Exp4–Exp6).

TABLE I

AN OVERVIEW OF ALL THE 6 EXPERIMENTS

EXPERIMENT	CONFIGURATION
EXP 1	Baseline(number: 1000)
EXP 2	Traditional Augmentation Only(1000)
EXP 3	Point-E Augmentation Only(1000)
EXP 4	Baseline(1000)& Traditional Augmentation(1000)
EXP 5	Baseline(1000) & Point-E Augmentation(1000)
EXP 6	Baseline(1000)&Traditional(1000) & Point-E(1000)

1) Traditional Augmentation

In this setting, each point cloud in the training set was transformed via a sequence of stochastic geometric perturbations. Each point cloud was rotated about the vertical axis by an angle $\theta \sim \mathcal{U}[-\pi, \pi]$, using the rotation matrix $R_y(\theta) \in \mathbb{R}^{3 \times 3}$. Uniform scaling was applied by multiplying all points with a random factor $s \sim \mathcal{U}[0.8, 1.2]$. Gaussian noise with standard deviation $\sigma = 0.01$ was added independently to each coordinate of each point, simulating sensor-level perturbation.

All augmentations preserved class identity and were applied during preprocessing to generate an enhanced training dataset. AIGC-Based Generative Augmentation (Point-E).

2) Generative Augmentation via Diffusion Model

A second category of augmentation was derived from synthetic point clouds generated using Point-E, a diffusion-based 3-D generative model. For each class in the dataset, 25 synthetic point clouds were generated using textual prompts, converted to meshes, and subsequently sampled to 1024 points per object. These synthetic samples were also normalized using the same centering and scaling strategy described for the test dataset.

This process produced a synthetic dataset parallel in structure and size to the real dataset, enabling evaluation of generative augmentation both independently (Exp3) and in conjunction with real data (Exp5, Exp6).

C. Network Architecture

The classification model used throughout the experiments was a PyTorch-based implementation of PointNet, an architecture specifically designed to operate on unordered point sets. The model receives a point cloud $P \in \mathbb{R}^{N \times 3}$. An input spatial transformer network (T-Net) learns a 3×3 affine matrix T_1 and applies it to each point

$$\tilde{p}'_i = T_1 \cdot p_i \quad (13)$$

The transformed points are passed through a series of shared multilayer perceptrons (MLPs) to produce local features $h_i \in$

$\mathbb{R}^d$. An optional feature transformation network (second T-Net) learns another matrix $T_2 \in \mathbb{R}^{d \times d}$, applied to h_i . To prevent degenerate transformations, an orthogonality regularization term is applied:

$$\mathcal{L}_{reg} = \|I - T_2 \cdot T_2^T\|_F^2 \quad (14)$$

A symmetric function (max pooling) aggregates local features into a global feature vector $g = \min_{i=1, \dots, N} h_i$. The global feature vector is passed through fully connected layers with dropout and ReLU activations. The final prediction $\hat{y} \in \mathbb{R}^c$ is produced via a softmax activation, and the model is trained using the cross-entropy loss as,

$$\mathcal{L}_{CE} = - \sum_{c=1}^C y_c \log(\hat{y}_c) \quad (15)$$

The total training objective combines classification loss and regularization as,

$$\mathcal{L} = \mathcal{L}_{CE} + \lambda \cdot \mathcal{L}_{reg} \quad (16)$$

Where λ is a fixed regularization coefficient set to 0.001.

D. Training Protocol and Evaluation

Each of the six experiment configurations was trained independently for 50 epochs using the Adam optimizer with a learning rate of 0.001 and a batch size of 32. Training was conducted on a single GPU device. All model weights were initialized uniformly, and dropout layers with a probability of 0.3 were included to reduce overfitting.

During training, epoch-wise loss and accuracy were recorded to CSV log files for subsequent analysis. All training was conducted using a modular pipeline script that automatically ingested the dataset structure, applied normalization, instantiated the model, and tracked performance metrics.

After training, all models were evaluated on the common 1,000-sample test set. Evaluation was performed using a dedicated inference script that loads each model checkpoint, feeds the test samples in batch mode, and records predicted class labels. Two metrics were computed: overall accuracy and macro-averaged F1-score, using standard definitions from the scikit-learn library.

Performance summaries were stored in CSV files and automatically parsed using an analysis script to generate comparative bar plots and summary tables. These visualizations were used to quantify the impact of each augmentation strategy on classification performance and to highlight generalization trends.

V. RESULTS AND DISCUSSION

To assess the impact of augmentation strategies on model performance, all six trained models (Exp1–Exp6) were evaluated on the standardized 1,000-sample test set constructed from the ModelNet40 dataset. Table I summarizes the classification accuracy and macro-averaged F1-scores for each experimental configuration.

TABLE II

TEST PERFORMANCE UNDER DIFFERENT AUGMENTATION CONDITIONS

EXPERIMENT	TRAINING CONFIGURATION	ACCURACY (%)	F1-SCORE (%)
EXP1	Original data only	87.4	86.7
EXP2	Traditional augmentation only	90.2	89.4

EXP3	AIGC-based augmentation only (Point-E)	88.6	87.9
EXP4	Original & Traditional augmentation	91.7	91.1
EXP5	Original & AIGC-based augmentation	90.6	89.9
EXP6	Combined augmentation (all sources)	92.6	91.9

The baseline model (Exp1) trained without augmentation achieved 87.4% accuracy and 86.7% F1-score, serving as a reference point for performance comparisons. Introducing traditional augmentation alone (Exp2) resulted in a notable improvement of 2.8% in accuracy and 2.7% in F1-score, indicating that geometric perturbations such as rotation, jittering, and scaling effectively enhance the network's ability to generalize to unseen orientations and spatial configurations. The use of AIGC-generated point clouds in isolation (Exp3) provided a moderate improvement over the baseline, achieving 88.6% accuracy. Although this suggests that Point-E-based samples are broadly consistent with class semantics, their benefit appears more limited than that of traditional transformations. This is likely due to generative imperfections, such as shape distortions or semantic ambiguity, which may introduce latent noise into the training distribution.

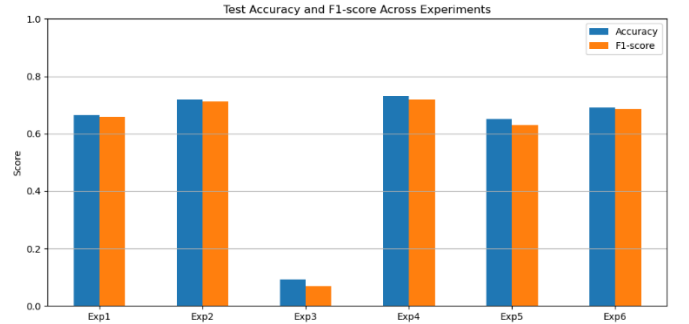


Fig. 1. The Test Accuracy and F1-score Across Experiments.

Combining original data with traditional augmentation (Exp4) yielded a further increase in performance, reaching 91.7% accuracy and 91.1% F1-score. When Point-E-generated data were added to the baseline (Exp5), a comparable improvement was observed, although not as pronounced as in Exp4. This implies that generative augmentation is capable of enhancing diversity, particularly in underrepresented regions of the feature space.

The best overall performance was achieved in Exp6, where traditional and AIGC augmentations were jointly incorporated alongside the original dataset. This configuration attained an accuracy of 92.6% and an F1-score of 91.9%, demonstrating that the two augmentation modalities may provide complementary benefits when balanced appropriately.

Training loss and accuracy trajectories, visualized using epoch-wise logs, confirm that augmented models (Exp2–Exp6) converge more quickly and achieve lower final loss values than the baseline. In particular, Exp4 and Exp6 showed both rapid initial convergence and stable plateauing, suggesting that the added data complexity from augmentations improved both

gradient flow and robustness during optimization. In contrast, Exp3 showed higher variance in loss, which may reflect inconsistencies in the generative data.

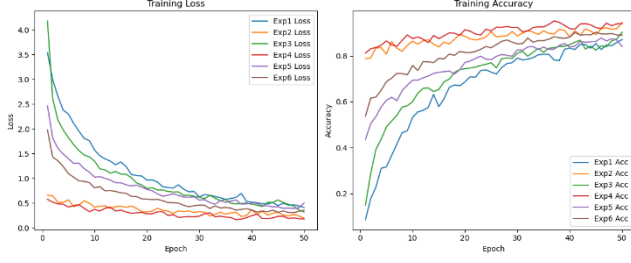


Fig. 2. The Training loss and Accuracy of Experiment 1-6.

Visual inspection of the Point-E-generated samples revealed that while many shapes exhibited class-relevant global geometry, a subset contained structural defects, disconnected regions, or topological noise. These imperfections may explain the marginally lower performance of generative augmentation compared to traditional methods when used alone. Nevertheless, when incorporated in combination (Exp6), such imperfections appear to act as a form of regularization, expanding the model’s hypothesis space and reducing overfitting to specific training patterns.

These findings underscore the importance of data-centric approaches in 3-D deep learning. While traditional geometric augmentation remains a reliable and computationally efficient method for enhancing model generalization, generative augmentation—despite its current limitations—demonstrates potential for contributing additional diversity to the training distribution. The key lies in balancing real and synthetic data to harness the advantages of each without overwhelming the learner with noisy or ambiguous inputs.

V. CONCLUSION

This study presented a systematic comparison between traditional and AIGC-based data augmentation strategies for 3-D point cloud classification using the PointNet architecture. Through six controlled experiments on the ModelNet40 dataset, we demonstrated that both augmentation paradigms improve classification performance relative to training on original data alone. Traditional geometric augmentation yielded the most substantial gains when applied independently, while Point-E-based synthetic samples provided moderate improvements. The highest accuracy and F1-score were achieved when all data sources were combined, suggesting that generative augmentation can serve as a valuable complement to conventional techniques.

Our findings highlight the practical value of integrating diverse data augmentation schemes in training pipelines, particularly in data-limited scenarios. While current generative models such as Point-E exhibit imperfections, their role in augmenting long-tail or structurally underrepresented classes is promising. Future work may focus on refining the fidelity and controllability of 3-D generative models, and on designing curriculum learning or data filtering mechanisms to better integrate synthetic data into supervised learning workflows.

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APPENDIX

Personal and Group Reflection

Junseok Kim

During this project, I was mainly responsible for building the baseline dataset, implementing data augmentation using Point-E, and designing the overall project structure. These roles allowed me to gain a deeper understanding of both data preparation strategies and the practical challenges of integrating generative models into a 3D classification pipeline.

One of the key difficulties I faced was dealing with the quality issues of Point-E generated point clouds. While this diffusion-based model is capable of generating diverse 3D shapes, many samples exhibited structural inconsistencies or semantic ambiguities, which negatively affected model performance. To overcome this, I developed filtering criteria to remove low-quality samples and carefully balanced the use of generative data alongside the original dataset.

Another challenge was establishing an effective project structure that enabled smooth collaboration and experimentation. I took the lead in defining our experimental configurations and ensuring that our baseline, traditional augmentation, and AIGC-based augmentation experiments were systematically organized. This helped streamline the implementation process and avoid redundant efforts among team members.

Through this project, I enhanced my practical skills in 3D data processing, diffusion models, and project management. I also learned the importance of critically evaluating data quality and understanding the limitations of cutting-edge generative

models before applying them in real-world tasks. This experience strengthened my ability to balance theoretical knowledge with hands-on experimentation, which will be valuable for future research.

Yekai

During this project, my primary responsibilities were data preprocessing, PointNet model training, and the management of datasets for experiments involving original, traditionally augmented, and Point-E-generated point clouds. One major challenge was maintaining dataset integrity and correctly balancing the class samples for different experiments, which required automated sanity checks and dynamic regeneration of train_files.txt to ensure consistency. Additionally, adapting the training pipeline from training-from-scratch to fine-tuning a pretrained model initially led to weight-loading issues, which I resolved by standardizing checkpoint loading and carefully managing layer freezing. Optimizing computational efficiency was also essential—I integrated mixed-precision training and cached intermediate features, reducing GPU memory usage significantly and enabling efficient overnight experiments. Through iterative collaboration and responding quickly to team feedback, I improved my skills in robust pipeline engineering, data quality control, and reproducible experimentation.

Xujiaheng

During the PointNet experiment evaluation project, I encountered several challenges that contributed to my technical and project management growth. Handling the large ModelNet40 dataset was a significant hurdle, as normalizing and resampling thousands of point clouds to meet specific requirements was time-consuming and required attention to detail. Additionally, evaluating six different models, each with unique fine-tuning strategies, demanded consistency in methodology to ensure fair comparisons. Generating meaningful visualizations from the training logs and accurately plotting accuracy and loss curves also presented difficulties, as I had to ensure the results were clear and insightful, highlighting trends like convergence and potential overfitting. The project also helped me develop better time management skills, as I had to balance tasks such as dataset preparation, model evaluation, and result visualization efficiently. Despite these challenges, the experience allowed me to enhance my technical abilities, improve my analytical thinking, and strengthen my ability to manage complex tasks under tight deadlines.

Gaohe

As the primary contributor responsible for drafting and structuring this project report, I have gained valuable experience in organizing technical content, interpreting experimental outcomes, and presenting complex concepts in a coherent academic format. Through analyzing augmentation strategies for 3-D point cloud classification, I have deepened my understanding of both traditional and generative data augmentation techniques, as well as the challenges involved in benchmarking learning systems. This project not only enhanced my technical writing and research synthesis skills, but also strengthened my ability to communicate machine learning workflows in a structured and impactful way. The process has

been instrumental in bridging the gap between theoretical knowledge and practical implementation in the domain of 3-D deep learning.

Group

Throughout this project, our team explored the comparative effectiveness of traditional data augmentation and AIGC-based augmentation for 3D point cloud classification using the PointNet architecture. While our technical focus remained on improving model generalization and accuracy, we also encountered and overcame various challenges as a team, gaining valuable insights into both collaborative research and advanced machine learning techniques.

The main challenge we faced was integrating AIGC-generated data using the Point-E model into our training pipeline. Many of the generated point clouds contained structural defects and semantic inconsistencies, which initially degraded model performance. To address this, we conducted multiple ablation studies and carefully filtered low-quality generated samples, allowing us to maintain data diversity while minimizing the negative impact of noisy data.

We also struggled with GPU resource limitations, which restricted the number of experiments we could conduct simultaneously. To overcome this, we carefully planned our experiment schedule, optimized our code for faster execution, and made efficient use of available hardware resources.

Through these challenges, we improved our technical proficiency in 3D deep learning, diffusion-based generative models, and experiment pipeline management. Most importantly, we developed strong collaboration and communication skills by effectively dividing tasks, sharing ideas, and supporting each other under tight deadlines.

This project has taught us the critical importance of both data quality and augmentation strategies in deep learning research. It also reinforced the value of teamwork, clear communication, and perseverance when working on complex projects under real-world constraints.

Contribution

Team Member	Role and Contribution	Percentage (%)
Junseok Kim	Project Idea Initiation & Overall Direction, Generative AI-based Data Generation (Point-E), Traditional Augmentation	36%
Yekai	Data Preprocessing, Dataset Preparation, Model Training	23%
Xu Jiaheng	Evaluation, Ablation Study, Performance Analysis	23%
He Gao	Final Report & Proposal Report Writing, Team Reflection Drafting, Project Introduction	18%